Run report: H4-memsweep-run-20261018T2209Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode memsweep (Addendum H4) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 6083c58aa45e04e4d3194689b3b3e6ef5c7be115
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 22:09:13 UTC 2026
host: this session's cloud container (L0)
cmd: n2_routes <dir> memsweep (N2_THREADS=4, sizes 32,36)
```

memsweep_summary.txt

```
H4 bits 32 theta 0.125: trials/p 0.7001 stored/p 0.00558 time*memory/p^2 0.00391 (40 runs)
H4 bits 32 theta 0.25: trials/p 0.6240 stored/p 0.02010 time*memory/p^2 0.01254 (40 runs)
H4 bits 32 theta 0.5: trials/p 0.5625 stored/p 0.07192 time*memory/p^2 0.04045 (40 runs)
H4 bits 36 theta 0.125: trials/p 0.6166 stored/p 0.00478 time*memory/p^2 0.00295 (40 runs)
H4 bits 36 theta 0.25: trials/p 0.5883 stored/p 0.01824 time*memory/p^2 0.01073 (40 runs)
H4 bits 36 theta 0.5: trials/p 0.4976 stored/p 0.06243 time*memory/p^2 0.03106 (40 runs)
H1/H4 all answers verified: true
```

memsweep.csv (first 12 of 240 rows)

bits	h	instance	p	ell	cofactor_exponent	target	arm	theta	steps	relations	stored_rows	verified	model_vertices	model_first_cycle_edges	M_total	seconds
32	4	0	73291	1342902107	2	0	Jtheta	0.125	70182	546	545	1	0	NaN	3363480	0.029289
32	4	0	73291	1342902107	2	0	Jtheta	0.25	70159	2218	2217	1	0	NaN	3362440	0.027032
32	4	0	73291	1342902107	2	0	Jtheta	0.5	61720	7949	7948	1	0	NaN	2975117	0.029596
32	4	0	73291	1342902107	2	1	Jtheta	0.125	29140	218	217	1	0	NaN	1477312	0.011393
32	4	0	73291	1342902107	2	1	Jtheta	0.25	29079	931	930	1	0	NaN	1474562	0.012031
32	4	0	73291	1342902107	2	1	Jtheta	0.5	20745	2631	2630	1	0	NaN	1090426	0.009932
32	4	0	73291	1342902107	2	2	Jtheta	0.125	77930	623	622	1	0	NaN	3718434	0.029318
32	4	0	73291	1342902107	2	2	Jtheta	0.25	77930	2508	2507	1	0	NaN	3718435	0.029932
32	4	0	73291	1342902107	2	2	Jtheta	0.5	22413	2830	2829	1	0	NaN	1167650	0.009705
32	4	0	73291	1342902107	2	3	Jtheta	0.125	41382	359	358	1	0	NaN	2037996	0.017888
32	4	0	73291	1342902107	2	3	Jtheta	0.25	38086	1266	1265	1	0	NaN	1885475	0.015026
32	4	0	73291	1342902107	2	3	Jtheta	0.5	59608	7550	7549	1	0	NaN	2871841	0.023823

Graph

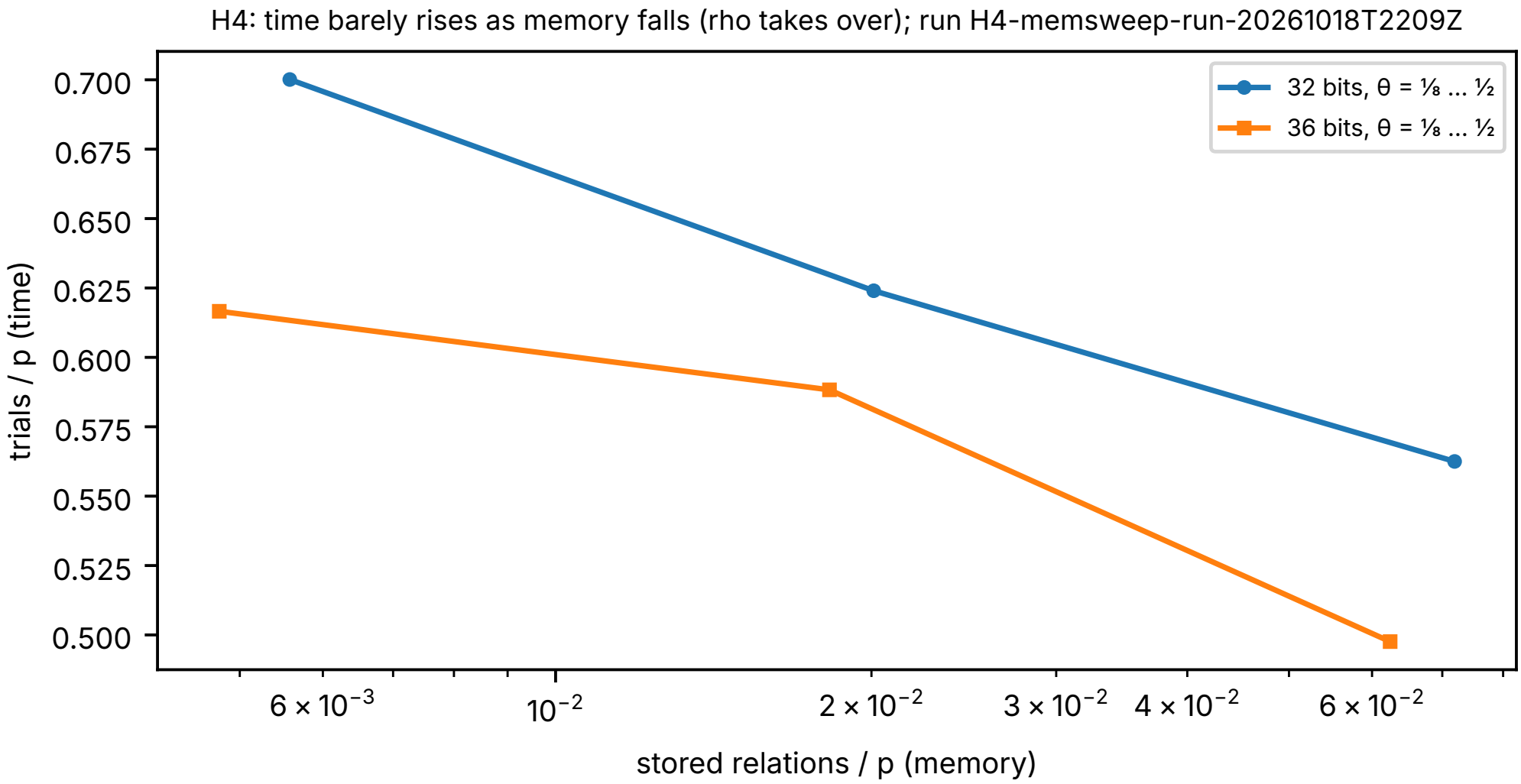


Figure 1: H4: trials against stored relations as the distinguished fraction theta falls.